Technical Service Bulletin

Technical Service Bulletin: TSB104266	**Released Date:** 22-Dec-2010
Alternator Change	

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This service parts topic announces a product change in Cummins-branded alternators. Delco Remy America (DRA), the supplier of the 21SI and 30SI Series alternators, has discontinued the production of these alternators. The 21SI will be replaced by the 22SI, and the 30SI will be replaced by the 33SI. This topic offers a comparison between the 21SI and 22SI and the 30SI and 33SI alternators.

The tables below summarize the engine serial number first for the 22SI and 33SI Series alternators. The tables are incomplete because the various engine plants are using up their stock of the 21SI and 30SI Series alternators before ordering the new 22SI and 33SI alternators. This service parts topic will be updated with the completed engine serial number first information as soon as it is available.

22SI ALTERNATOR ENGINE SERIAL NUMBER FIRST

22SI ALTERNATOR ENGINE SERIAL NUMBER FIRST				
	Engine Series			
Part No.	**B/C**	**L/M**	**N/Signature**	**K/V**
3935527	45822614	34950352		
3935528	45827094			N/A
3935529				
3935530	45824427	34951091		
3935531			N/A	N/A

Note : N/A indicates that parts are not available.

33SI ALTERNATOR ENGINE SERIAL NUMBER FIRST				
	Engine Series			
Part No.	B/C	L/M	N/Signature	K/V
4000590		34951754		
3400582	N/A	N/A		
4000589	N/A			
3400698	N/A	N/A		
3413092	N/A	N/A		N/A

Note : N/A indicates that parts are **not** available.

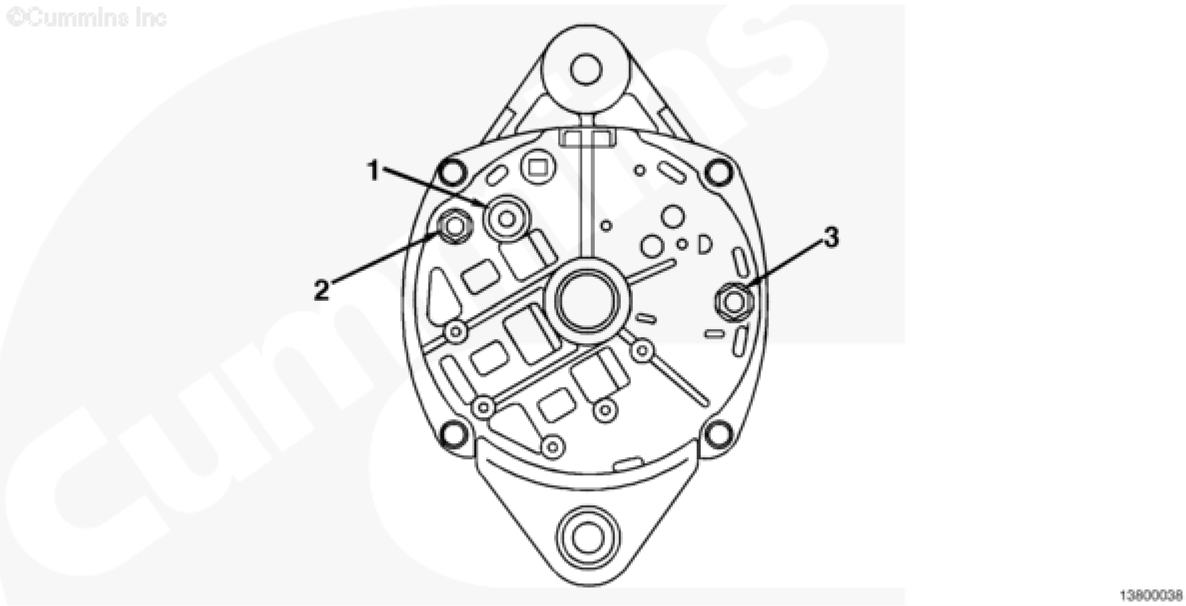

Figure 1, 21SI Alternator

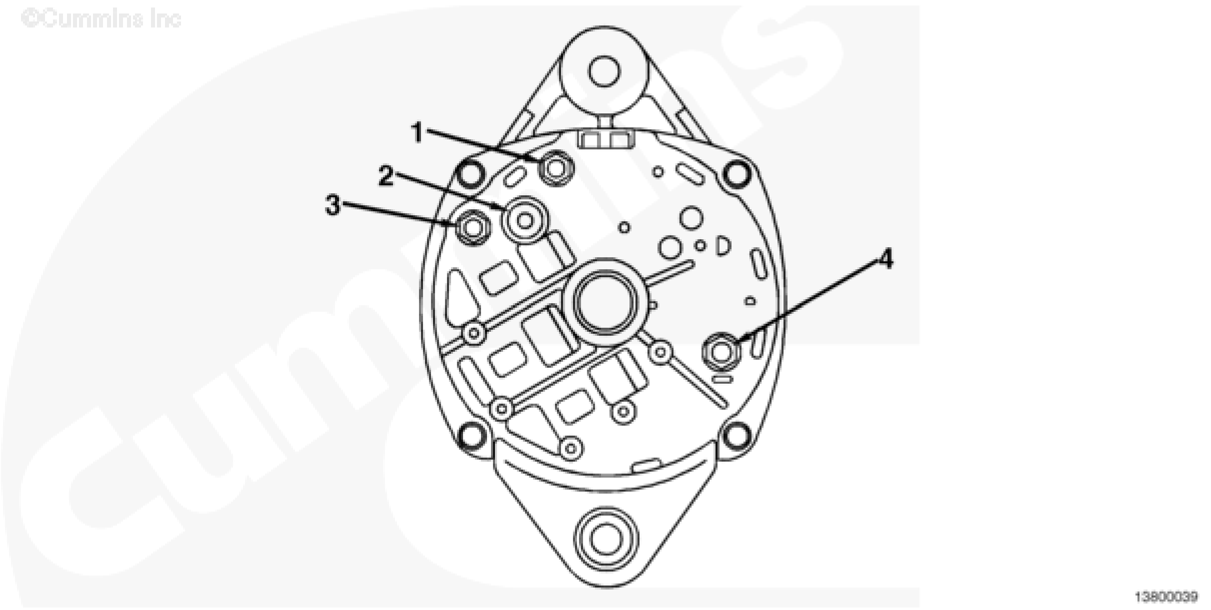

Figure 2, 22SI Alternator

21SI and 22SI Comparison

The 21SI and 22SI alternators are interchangeable on the same mount. There are two minor differences that can affect their fit in some applications: Ground terminal location and maximum housing width.

Ground Terminal Location

The ground terminal location of the 22SI alternator is 20 mm [0.79 in] lower and 11 mm [0.43 in] to the left of the 21SI ground terminal as viewed from the rear of the housing. (See Figs. 1 and 2.) The difference between the locations of the ground terminal on the two alternators is 23 mm [0.91 in].

Maximum Housing Width

The 22SI alternator is 5 mm [0.20 in] wider than the 21SI alternator. The maximum width of the 21SI alternator is 145 mm [5.71 in] compared to a width of 150 mm [5.91 in] for the 22SI.

Note : All 22SI alternators have an indicator (I) terminal and a relay (R) terminal. Use of these terminals is optional. An explanation of the I and R terminals is located at the end of this topic.

Figure 1, 21SI Alternator	
Reference No.	**Description**
1	Battery terminal
2	Relay terminal
3	Ground terminal

Figure 2, 22SI Alternator	
Reference No.	**Description**
1	Indicator terminal

Figure 2, 22SI Alternator	
Reference No.	**Description**
2	Battery terminal
3	Relay terminal
4	Ground terminal

In order to print this table properly, please set your printer paper orientation to "LANDSCAPE."

Table 1, 21SI and 22SI Feature Comparison

Cummins Part No.	Delco Part No.	Series	Rotation	VDC	Amps	G	S	Bat+	R Trm	I Trm	G Trm
3918558	19010194	21SI	**CW/ CCW**	12	100	N	1	1/4	PIN		1/4
3935527	**19020379**	**22SI**	**CW**	**12**	**100**	**N**	**1**	**1/4**	**PIN**	**No. 10**	**1/4**
3920615	19010196	21SI	**CW**	12	130	N	1	1/4	PIN		1/4
3935528	**19020380**	**22SI**	**CW**	**12**	**130**	**N**	**1**	**1/4**	**PIN**	**No. 10**	**1/4**
3920616	19010202	21SI	**CW/ CCW**	12	160	N	1	1/4	PIN		1/4
3925529	**19020381**	**22SI**	**CW**	**12**	**160**	**N**	**1**	**1/4**	**PIN**	**No. 10**	**1/4**
3920617	19010199	21SI	**CW/ CCW**	24	50	N	1	M6	M4	M4	1/4
3935531	**19020383**	**22SI**	**CW**	**24**	**50**	**N**	**1**	**M6**	**M4**	**M4**	**1/4**
3920618	19010195	21SI	**CW/ CCW**	24	70	N	1	M6	M4	M4	1/4
3935530	**19020382**	**22SI**	**CW**	**24**	**70**	**N**	**1**	**M6**	**M4**	**M4**	**1/4**
3679712	**19020314**	**22SI**	**CW**	**12**	**130**	**N**	**1**	**5/16**	**No. 10**	**No. 10**	**1/4**

Table 1, 21SI and 22SI Feature Comparison											
Cummins Part No.	**Delco Part No.**	**Series**	**Rotation**	**VDC**	**Amps**	**G**	**S**	**Bat+**	**R Trm**	**I Trm**	**G Trm**
4003445	**19020375**	**22SI**	**CW**	**12**	**130**	**N**	**3**	**M6**	**M4**	**BLADE**	**1/4**
4003446	**19020376**	**22SI**	**CW**	**24**	**70**	**N**	**3**	**M6**	**M4**	**BLADE**	**1/4**

Table Key					
S	=	System (1-wire, 3-wire	**I Trm**	=	Indicator Terminal Size
Bat+	=	Alternator Output Terminal Size	**G Trm**	=	Ground Terminal Size
VDC	=	Voltage	**G**	=	Ground Type
Amps	=	Amperage	**N**	=	Frame Ground
R Trm	=	Relay Terminal Size	**I**	=	Insulated

Table 2, 22SI Options						
Engine Series	**Old Option**	**New Option**	**Old Part No.**	**New Part No.**	**VDC**	**Amps**
B/C	EE9055	EE9124	3918558	3935527	12	100
	EE9056	EE9125	3920615	3935528	12	130
	EE9057	EE9126	3920616	3935529	12	160
	EE9058	EE9128	3920617	3935531	24	50
	EE9059	EE9127	3920618	3935530	24	70
	EE9067	EE9129	3918558	3935527	12	100
	EE9073	EE9130	3920615	3935528	12	130
	EE9074	EE9131	3920616	3935529	12	160
	EE9075	EE9133	3920617	3935531	24	50
	EE9076	EE9132	3920618	3935530	24	70
	EE9084	EE9137	3920618	3935530	24	70

Table 2, 22SI Options						
Engine Series	**Old Option**	**New Option**	**Old Part No.**	**New Part No.**	**VDC**	**Amps**
L10	EE8015	EE2044	3920618	3935530	24	70
	EE8029	EE2041	3918558	3935527	12	100
	EE8131	EE2041	3918558	3935527	12	100
	EE8132	EE2043	3920616	3935529	12	160
	EE8133	EE2044	3920618	3935530	24	70
	EE8140	EE2042	3920615	3935528	12	130
M11/ISM	EE2005	EE2047	3918558	3935527	12	100
	EE20056	EE2049	3920616	3935529	12	160
	EE2007	EE2050	3920618	3935530	24	70
	EE2011	EE2048	3920615	3935528	12	130
	EE2028	EE2051	3920617	3935531	24	50
N/NT/N14 / Signature	EE1193	EE1244	3918588	3935527	12	100
	EE1209	EE1245	3920615	3935528	12	130
	EE1212	EE1248	3920618	3935530	24	70
	EE8131	EE1254	3918558	3935527	12	100
	EE8132	EE1255	3920616	3935529	12	160
	EE8133	EE1256	3920618	3935530	24	70
V903	EE8013	EE3125	3918558	3935527	12	100
	EE8015	EE3126	3920618	3935530	24	70
QSK19	EE4032	EE4063	3920618	3935530	24	70
	EE4035	EE4065	3920616	3935529	12	160
	EE8013	EE4061	3918558	3935527	12	100
	EE8015	EE4064	3920618	3935530	24	70
	EE8029	EE4061	3918558	3935527	12	100
	EE8131	EE4061	3918558	3935527	12	100
	EE8132	EE4062	3920616	3935529	12	160
	EE8133	EE4064	3920618	3935530	24	70

Table 2, 22SI Options						
Engine Series	**Old Option**	**New Option**	**Old Part No.**	**New Part No.**	**VDC**	**Amps**
KV/QSKV	EE8015	EE6055	3920618	3935530	24	70
	EE8029	EE6052	3918558	3935527	12	100
	EE8131	EE6053	3918558	3935527	12	100
	EE8132	EE6054	3920616	3935529	12	160
	EE8133	EE6056	3920618	3935530	24	70

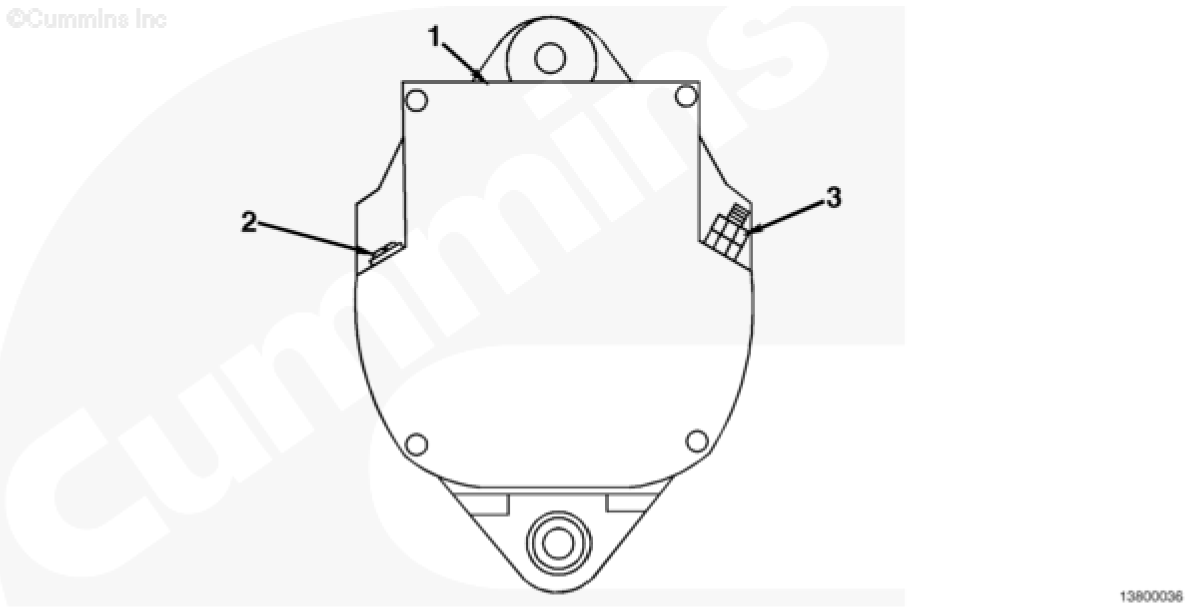

Figure 3, 30SI Alternator

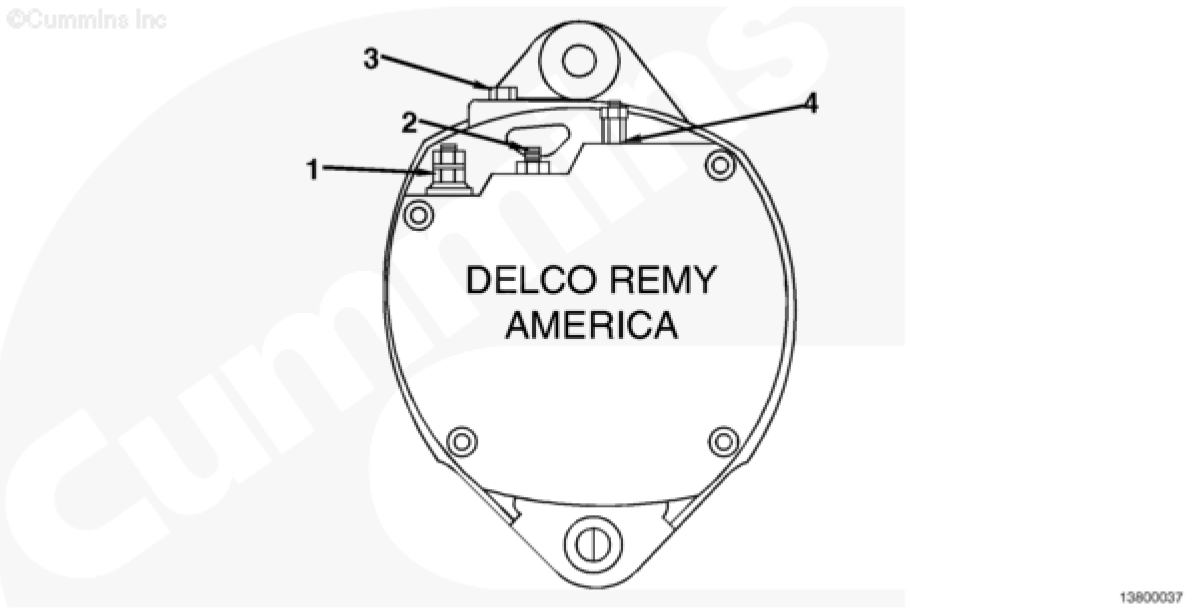

Figure 4, 33SI Alternator

30SI and 33SI Comparison

The 30SI and 33SI alternators are interchangeable on the same mount. The **only** differences that will affect installation are the terminal locations. (See Figs. 3 and 4.)

Ground Terminal Location

The ground terminal on both the 30SI and 33SI is located on top of the housing relative to the rear of the alternator.

Relay Terminal Location

The relay terminal location on the 30SI alternator is located on the left side when viewed from the rear of the housing. The relay terminal on the 33SI alternator is located in the upper left corner, next to the battery terminal, when viewed from the rear of the housing. The actual difference between the terminal locations is approximately 90 mm [3.54 in].

Battery Terminal Location

The battery terminal on the 30SI alternator is located on the right side when viewed from the rear of the housing. The battery terminal on the 33SI is located in the upper left corner of the alternator when viewed from the rear of the housing. The actual difference between the terminal locations is approximately 190 mm [7.48 in].

Indicator Light Terminal Location

The 30SI alternator does **not** have an indicator terminal. The 33SI indicator terminal is located in the upper middle portion of the rear housing, next to the relay terminal.

Note : Use of the relay and the indicator terminals is optional. An explanation of these terminals is located at the end of this topic.

Figure 3, 30SI Alternator	
Reference No.	**Description**
1	Ground terminal
2	Relay terminal
3	Battery terminal

Figure 4, 33SI Alternator	
Reference No.	**Description**
1	Battery terminal
2	Relay terminal
3	Ground terminal
4	Indicator terminal

In order to print this table properly, please set your printer paper orientation to "LANDSCAPE."

Table 3, 30SI and 33SI Feature Comparison											
Cummins Part No.	**Delco Part No.**	**Series**	**Rotation**	**VDC**	**Amps**	**G**	**S**	**Bat+**	**R Trm**	**I Trm**	**G Trm**
3008988	1117833	30SI/TR	**CW/CCW**	12/24	90	N	1	1/4	PIN		1/4
4000589	**19011193**	**33SI/TR**	**CW/CCW**	**12/24**	**110**	**N**	**1**	**1/4**	**PIN**	**No. 10**	**1/4**
3004806	1117839	30SI	**CW/CCW**	12	90	N	1	1/4	PIN		1/4
3400582	**19011174**	**33SI**	**CW/CCW**	**12**	**110**	**N**	**1**	**1/4**	**PIN**	**No. 10**	**1/4**
3000347	1117834	30SI	**CW/CCW**	24	60	I	1	1/4			
3056492	1117836	30SI	**CW/CCW**	24	75	N	1	1/4	PIN	No. 10	1/4
4000590	**19011192**	**33SI**	**CW/CCW**	**24**	**75**	**N**	**1**	**1/4**	**PIN**	**No. 10**	**1/4**
3072483	1117838	30SI	**CW/CCW**	24	100	N	1	1/4	PIN		1/4
3400698	**19011175**	**33SI**	**CW/CCW**	**24**	**100**	**N**	**1**	**1/4**	**PIN**	**No. 10**	**1/4**
3000346*	1117733	30SI	**CW/CCW**	32	60	I	1	1/4			1/4
3413092*	**1901116**	**33SI**	**CW/CCW**	**32**	**60**	**I**	**1**	**1/4**	**PIN**	**10**	
3680140	**19011173**	**33SI**	**CW/CCW**	**12**	**135**	**N**	**1**	**1/4**	**PIN**	**No. 10**	**1/4**

* Delco-branded alternator. All others are Cummins-branded.

Table Key					
S	=	System (1-wire, 3-wire	**I Trm**	=	Indicator Terminal Size

Table Key					
Bat+	=	Alternator Output Terminal Size	**G Trm**	=	Ground Terminal Size
VDC	=	Voltage	**G**	=	Ground Type
Amps	=	Amperage	**N**	=	Frame Ground
R Trm	=	Relay Terminal Size	**I**	=	Insulated

Note : All 30SI and 33SI alternators are marine corrosion protected.

Note : If one or two of the 30SI 24-VDC alternators are used on equipment with dual engines with dual alternators and a common battery pack, the customer will perhaps complain about the charge light staying on at engine idle. To prevent this, use **only** the 33SI 24-VDC alternators on both engines for dual-engine equipment with a common battery pack. The 33SI 24-VDC alternator has an improved voltage regulator to prevent the charge light from coming on at engine idle.

Table 4, 33SI Options

Engine Series	Old Option	New Option	Old Part No.	New Part No.	VDC	Amps
B/C	EE9038	EE9135	3056492	4000590	24	75
	EE9090	EE9034	3056492	4000590	24	75
L10	EE8044	EE2040	3000347	4000590	24	75
	EE8049	EE2031	3000347	3400582	12	110
	EE8068	EE2039	3008986	4000589	12/24	110
	EE8119	EE2040	3056492	4000590	24	75
M11/ISM	EE2009	EE2034	3004806	3400582	12	110
	EE2014	EE2046	3000347	4000590	24	75
	EE2016	EE2046	3056492	4000590	24	75
	EE2017	EE2045	3008988	4000589	12/24	110

Table 4, 33SI Options						
Engine Series	Old Option	New Option	Old Part No.	New Part No.	VDC	Amps
N/NT	EE8043	EE1249	3000346	3413092	32	60
	EE8044	EE1250	3000347	4000590	24	75
	EE8049	EE1248	3004806	3400582	12	110
	EE8068	EE1247	3008988	4000589	12/24	110
	EE8071	EE1251	3000347	4000590	24	75
	EE8072	EE1249	3000346	341309	32	60
	EE8119	EE1251	3056492	4000590	24	75
	EE8122	EE1248	3004806	3400582	12	110
	EE8129	EE1253	3072483	3400698	24	100
N14/ Signature	EE1194	EE1238	3004806	3400582	12	110
	EE1195	OBSOLETE				
	EE1210	EE1240	3056492	4000590	24	75
	EE1215	EE1242	3072483	3400698	24	100
	EE1217	EE1237	3008988	4000589	12/24	110
	EE1220	EE1243	3072483	3400698	24	100
	EE1232	EE1239	3000347	4000590	24	75
	EE1233	EE1240	3000347	4000590	24	75
KT/QSK19	EE4028	EE4053	3000347	4000590	24	75
	EE4031	EE4056	3056492	4000590	24	75
	EE4033	EE4054	3000347	4000590	24	75
	EE4034	EE4058	3072483	3400698	24	100
	EE4036	EE4050	3008988	4000588	12/24	110
	EE4040	EE4059	3072483	3400698	24	100
	EE8043	EE4067	3000346	3413092	32	60
	EE8044	EE4055	3000347	4000590	24	75
	EE8049	EE4052	3004806	3400582	12	110
	EE8068	EE4051	3008988	4000589	12/24	110
	EE8072	EE4068	3000346	3413092	32	60
	EE8119	EE4057	3056492	4000590	24	75
	EE8122	EE4052	3004806	3400582	12	110
	EE8129	EE4060	3072483	3400698	24	100

Table 4, 33SI Options						
Engine Series	**Old Option**	**New Option**	**Old Part No.**	**New Part No.**	**VDC**	**Amps**
QST30	EE5061	EE5067	3056492	4000590	24	75
	EE5062	EE5068	3072483	3400698	24	100
KV/QSKV	EE8044	EE6047	3000347	4000590	24	75
	EE8049	EE6048	3004806	3400082	12	110
	EE8068	EE6049	3008988	4000589	12/24	110
	EE8119	EE6050	3056492	4000590	24	75
	EE8129	EE6051	3072483	3400698	24	100

Indicator (I) Terminal

The main function of the indicator (I) terminal is to indicate if the alternator is working correctly. Typically, an indicator light is wired to this terminal. If the alternator is **not** charging properly, the light turns on. Another function of the indicator (I) terminal is that it can be used to supply up to 1 amp of output at system voltage.

Relay (R) Terminal

The function of the relay (R) terminal varies. It can supply up to 4 amps of output at one-half nominal alternator voltage to power items such as a tachometer or an hour meter.

One-Wire and Three-Wire Systems

One-Wire System

This is the simplest of the wiring systems because the **only** wires connected to the alternator are at the battery (BAT) and ground terminals. (See Fig. 5.) Connecting to the R terminal and I terminal is optional.

Three-Wire System

This system requires more wiring because it has a battery (BAT) terminal, R terminal, two blade terminals identified as No. 1 and No. 2, and a ground terminal. Typically, in the three-wire system, the No. 1 blade terminal serves as the I terminal. (See Fig. 5.) The advantage of the three-wire system is that it provides the same features as the one-wire system, plus remote sense. By connecting the No. 2 blade terminal to the battery's positive (+) terminal, the voltage is both sensed and regulated at the battery instead of the alternator. This eliminates the potential for voltage losses in the wiring from the alternator to the battery.

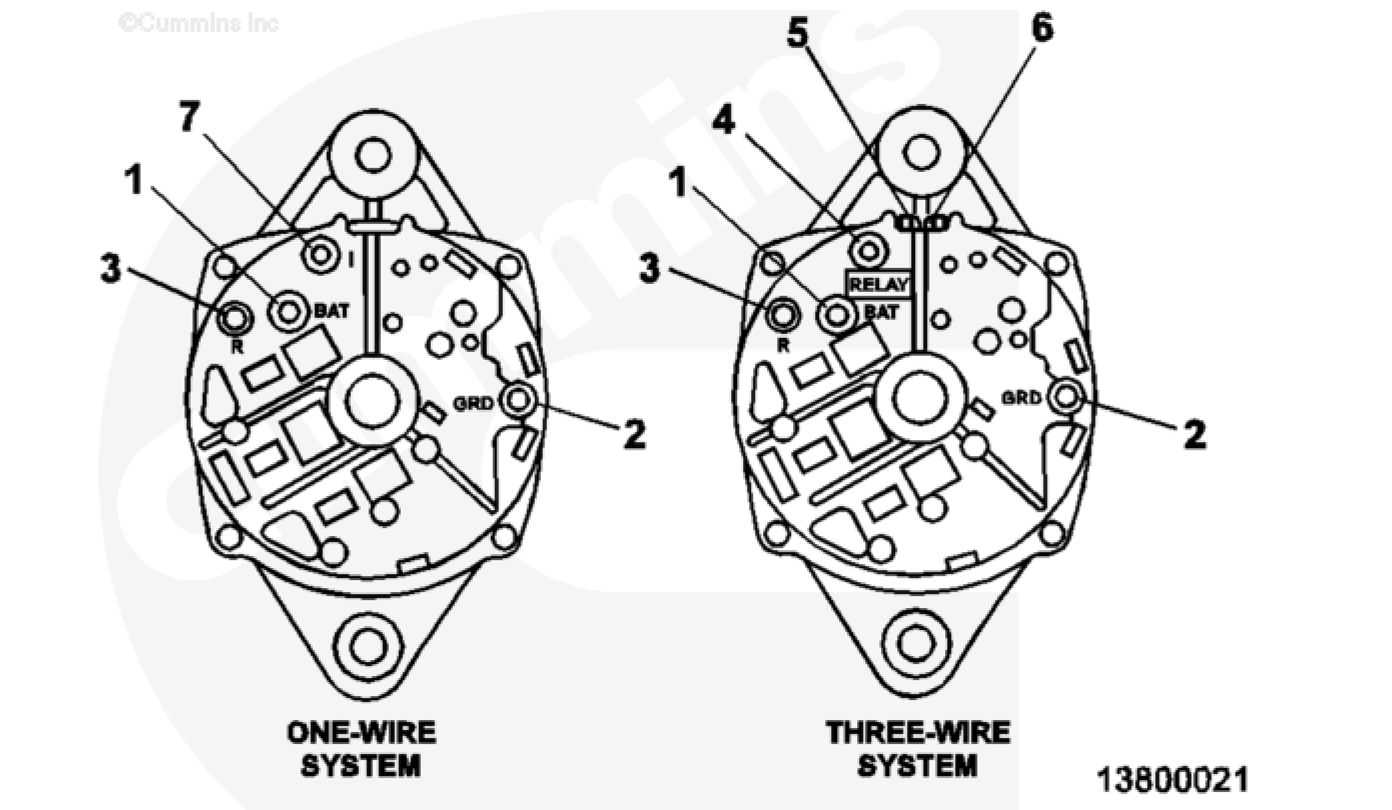

Fig. 5, Typical Alternator (Delco-Remy)

Key	Terminal	Connected to
1	Output or B	Battery
2	Ground* or B	Ground
3 and 4	Relay* or A.C.	Charge indicator, automatic lockout system, tachometer**
5	Terminal No. 1* or D+	Indicator light
6	Terminal No. 2	Voltage sense
7	Indicator*	Indicator light

* **Not** all alternators have this feature.

** Provides voltage pulses at about one-half system voltage at a frequency of one-tenth of generator rpm.

Document History

Date	Details
2010-12-20	Module Created